

A Global Dissipativity and Equilibrium-Stability Atlas for Lorenz–63

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Abstract

We close an all-parameter theorem for the Lorenz–63 equations rather than infer a parameter continuum from the familiar numerical attractor. For every $\sigma, \beta > 0$ and $\rho \in \mathbb{R}$, a shifted quadratic form gives an exact dissipation ledger, a global forward bound, and an explicit absorbing ellipsoid. We list every equilibrium, factor the origin spectrum, and reduce the symmetric wings’ cubic Routh–Hurwitz condition to one affine margin. This yields the exact stability split and, when $\sigma > \beta + 1$, the surface $\rho_H = \sigma(\sigma + \beta + 3)/(\sigma - \beta - 1)$ with its imaginary frequency. Zero σ , zero β , their intersection, and the $\rho = 1$ merger are resolved as separate singular faces. Only equilibrium and linear stability are claimed: neither nonlinear Hopf criticality nor universal chaos above ρ_H is inferred. Independent exact reconstruction, symbolic checks, canonical replay, and hostile mutations audit the result.

1 Convention and global dissipativity

We freeze the Lorenz–63 system

$$\dot{x} = \sigma(y - x), \quad \dot{y} = x(\rho - z) - y, \quad \dot{z} = xy - \beta z, \quad (1)$$

on $\mathbb{R}^3$, with $\sigma, \beta > 0$, $\rho \in \mathbb{R}$, and physical time t . It is equivariant under $(x, y, z) \mapsto (-x, -y, z)$, and its divergence is $-(\sigma + \beta + 1)$. Volume contraction alone will not be used as a dissipativity proof.

Theorem 1 (Explicit absorbing ellipsoid). *Let $c = \rho + \sigma$,*

$$V = x^2 + y^2 + (z - c)^2, \quad \kappa = \min\{2\sigma, 2, \beta\}.$$

Every solution of (1) is global forward and

$$\dot{V} = -2\sigma x^2 - 2y^2 - \beta z^2 - \beta(z - c)^2 + \beta c^2 \quad (2)$$

$$\leq -\kappa V + \beta c^2, \quad (3)$$

$$V(t) \leq e^{-\kappa t} V(0) + \frac{\beta c^2}{\kappa} (1 - e^{-\kappa t}). \quad (4)$$

Hence every ellipsoid $V \leq R$ with $R > \beta c^2/\kappa$ absorbs every trajectory.

Proof. Differentiate V along (1). The xyz terms cancel, while the remaining xy coefficient vanishes because $c = \rho + \sigma$. The last term is closed by

$$-2\beta z(z - c) = -\beta z^2 - \beta(z - c)^2 + \beta c^2,$$

which proves (2) and (3). Gronwall’s inequality gives (4). The vector field is polynomial, so a maximal finite-time solution could fail only by escaping every bounded set; (4) excludes that. On $V = R > \beta c^2/\kappa$, (3) points strictly inward, and (4) gives eventual entry. $\square$

2 Equilibria and exact characteristic polynomials

Theorem 2 (Complete positive-parameter atlas). *The origin O exists for all ρ and has*

$$\chi_O(\ell) = (\ell + \beta)[\ell^2 + (\sigma + 1)\ell + \sigma(1 - \rho)]. \quad (5)$$

It is asymptotically stable for $\rho < 1$, nonhyperbolic at $\rho = 1$, and a saddle for $\rho > 1$. If and only if $\rho > 1$, there are two further equilibria

$$E_{\pm} = (\pm\sqrt{\beta(\rho - 1)}, \pm\sqrt{\beta(\rho - 1)}, \rho - 1), \quad (6)$$

with common polynomial

$$\chi_E(\ell) = \ell^3 + (\sigma + \beta + 1)\ell^2 + \beta(\sigma + \rho)\ell + 2\sigma\beta(\rho - 1). \quad (7)$$

Proof. The equilibrium equations give $y = x$, $xy = \beta z$, and $x(\rho - z - 1) = 0$, proving the list. Substitution in

$$Df = \begin{pmatrix} -\sigma & \sigma & 0 \\ \rho - z & -1 & -x \\ y & x & -\beta \end{pmatrix}$$

gives (5) and (7). The quadratic Routh criterion in (5) gives the origin classification. $\square$

References

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Declarations

Scope. NO_BAD_EULER_OR_ROOT_NUMBER; no target arithmetic, target-zero, global-chaos, or Hilbert–Pólya claim. **Data and code.** The exact ledger and all audit programs accompany HCS-C227. **AI-use disclosure.** Generative tools assisted drafting and code generation; the released internal checks validate the displayed identities and metadata. This is not external peer review.